

IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

FALCUTÉ DES SCIENCES ET TECHNOLOGIES

(FST)

TROISIÈME ANNÉE

Rapport du travail de Laboratoire NO 3

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LE 8 /11/ 2026

Adressage IPv4 et IPv6 avec Cisco Packet Tracer

TD 3

L’objectif de ce TD est de :

1. Savoir attribuer des adresses IP valides aux machines.

2. Comprendre l’adressage IPv4 et IPv6.

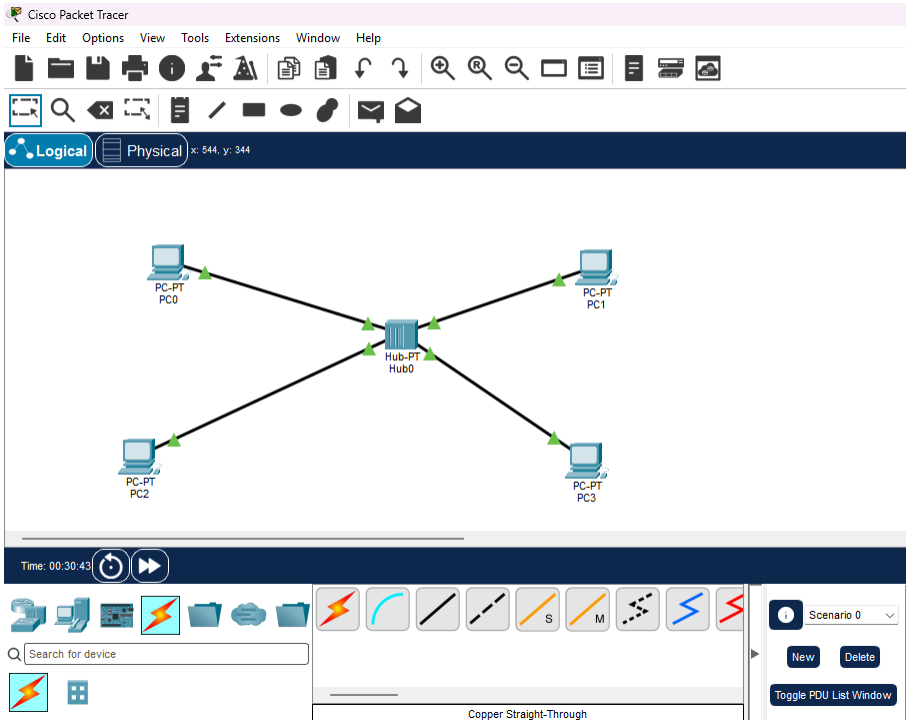
3. Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.

4. Vérifier la connectivité avec les commandes ping et ping ipv6.

Créez une topologie avec :

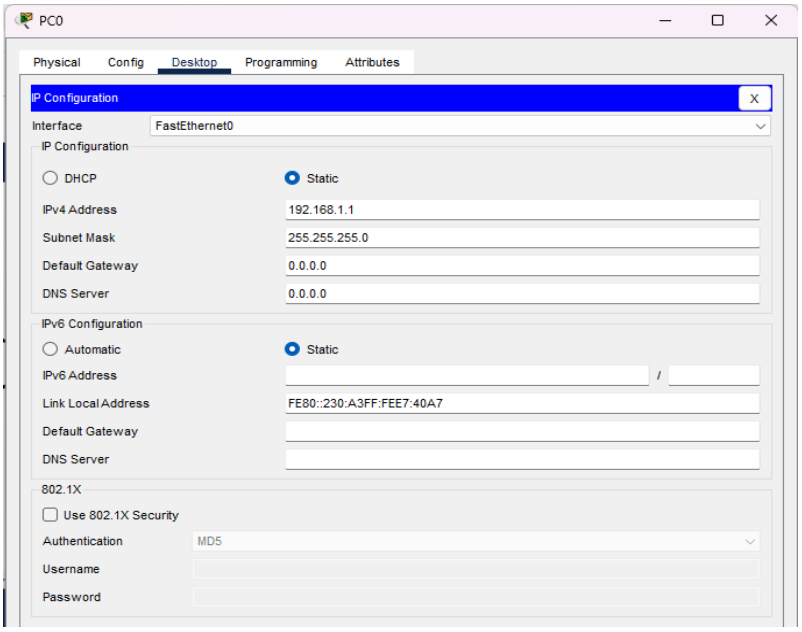
Un Hub.

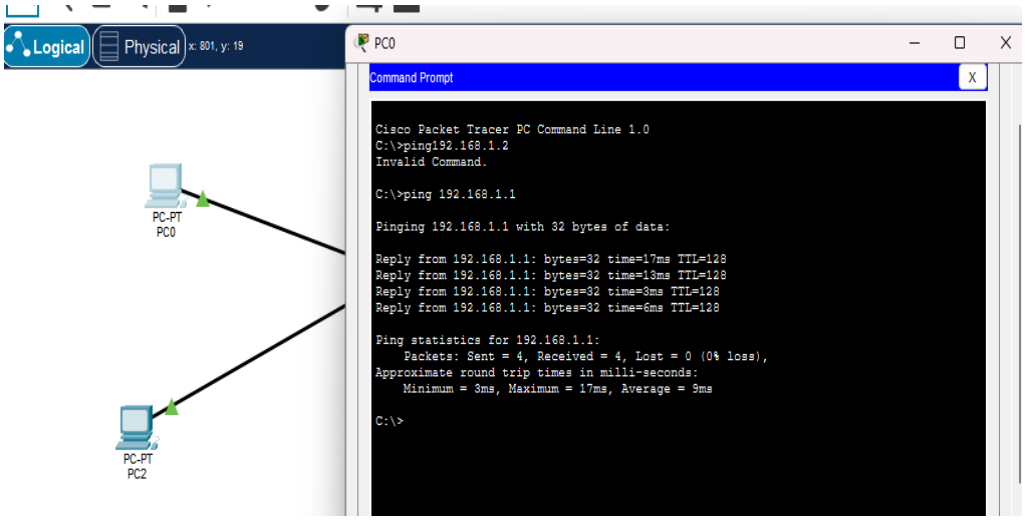
4 PC connectés au switch.



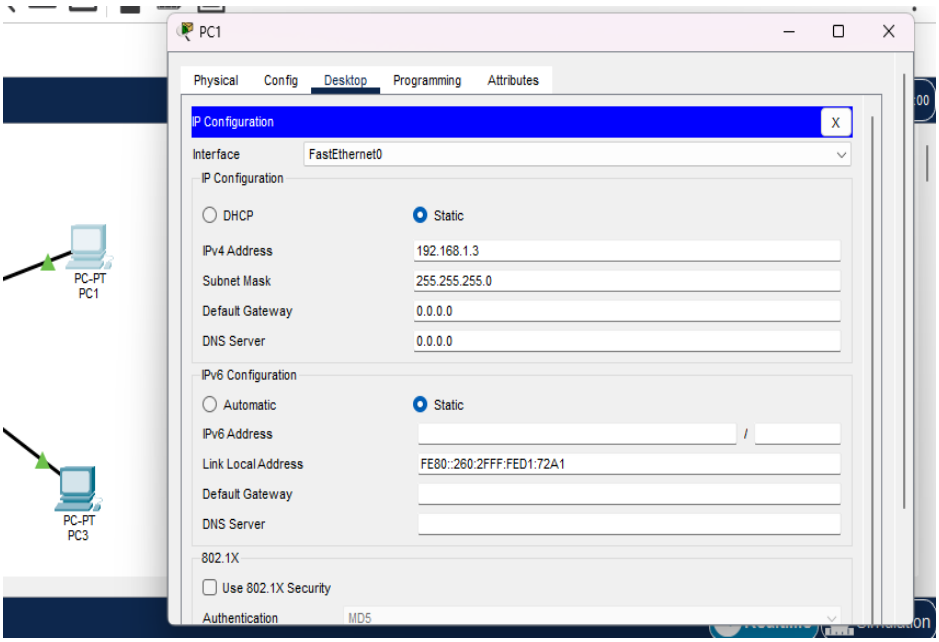
Configurez les adresses IPv4 suivantes :

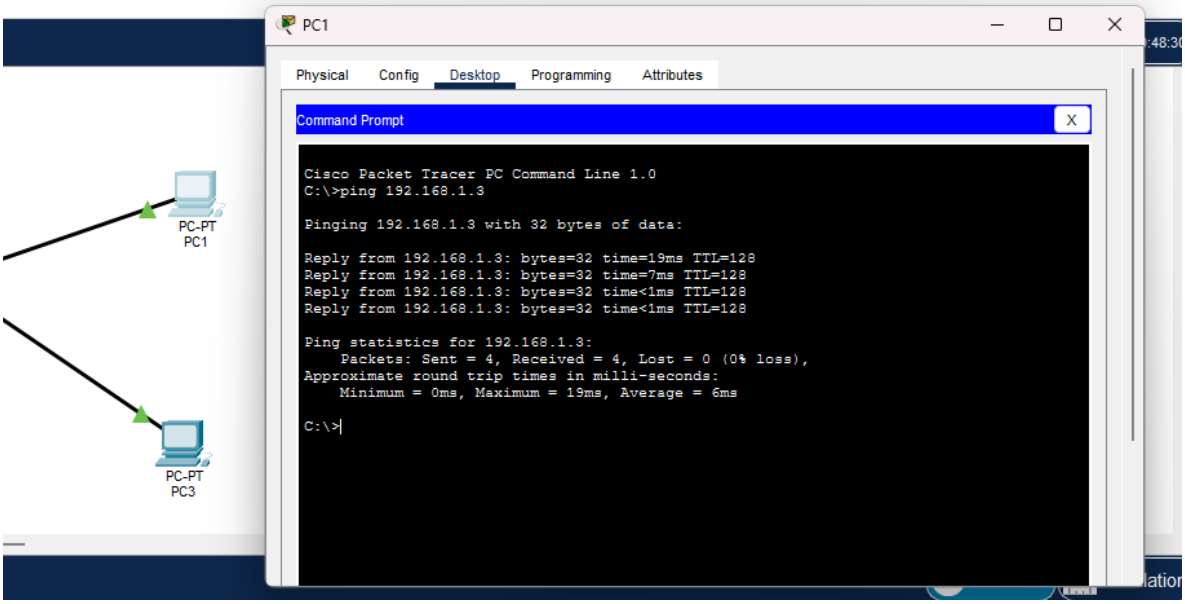
Pc : 0



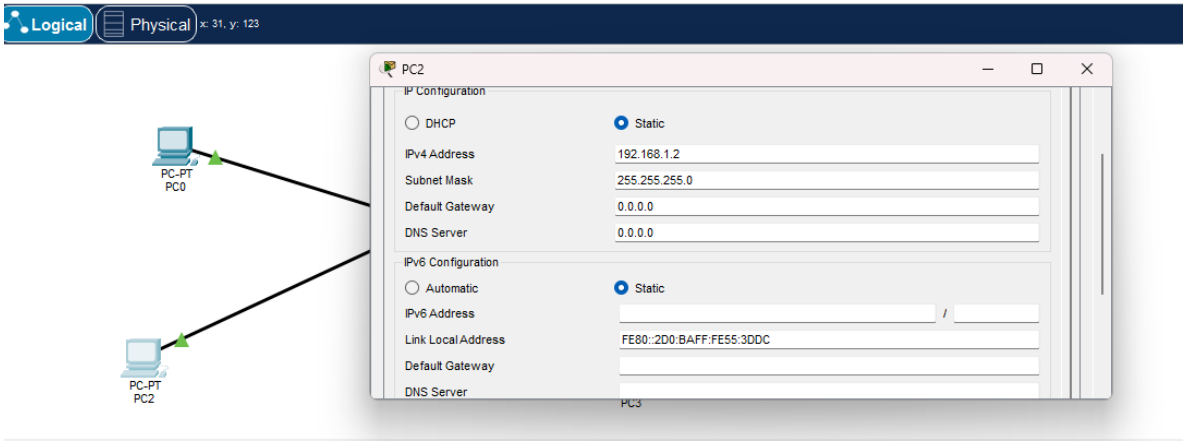


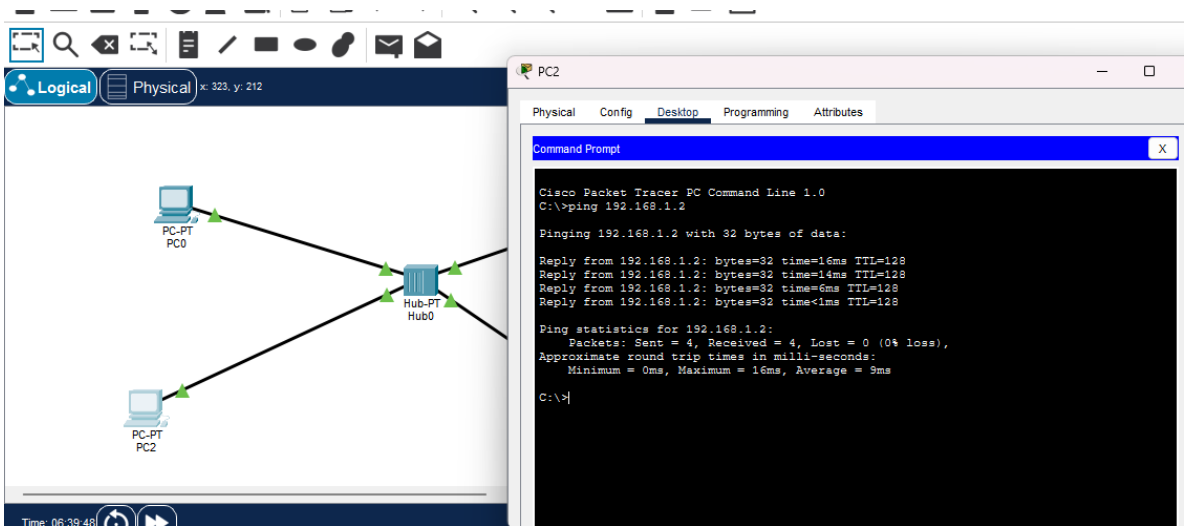
Pc 1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:2FFF:FED1:72A1
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
PC-PT
PC1
PC-PT
PC3

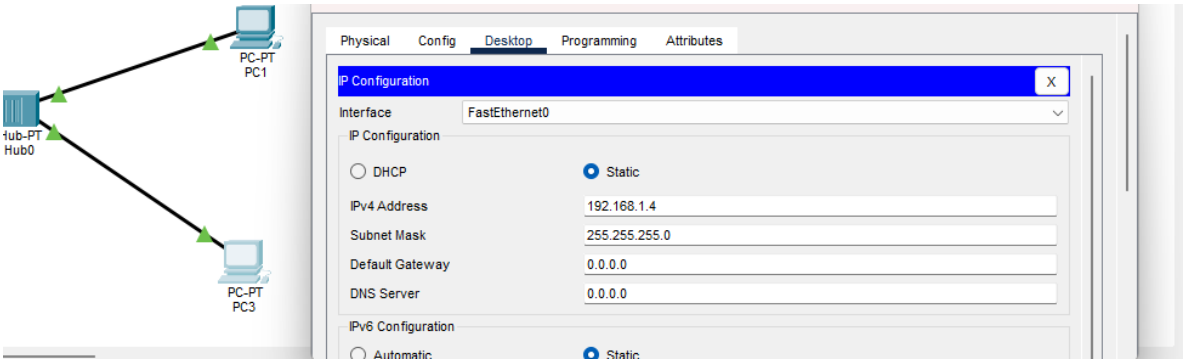

PC1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=19ms TTL=128
Reply from 192.168.1.3: bytes=32 time=7ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 19ms, Average = 6ms
C:\>
PC-PT
PC1
PC-PT
PC3

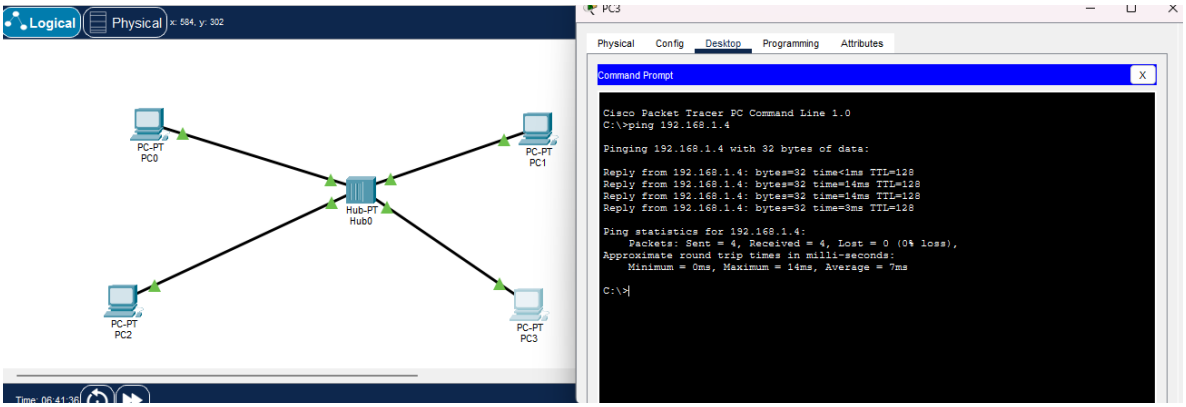
pc 2


Logical
Physical
x: 31, y: 123
PC2
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:BAFF:FE55:3DDC
Default Gateway
DNS Server
PC-PT
PC0
PC-PT
PC2
PC3

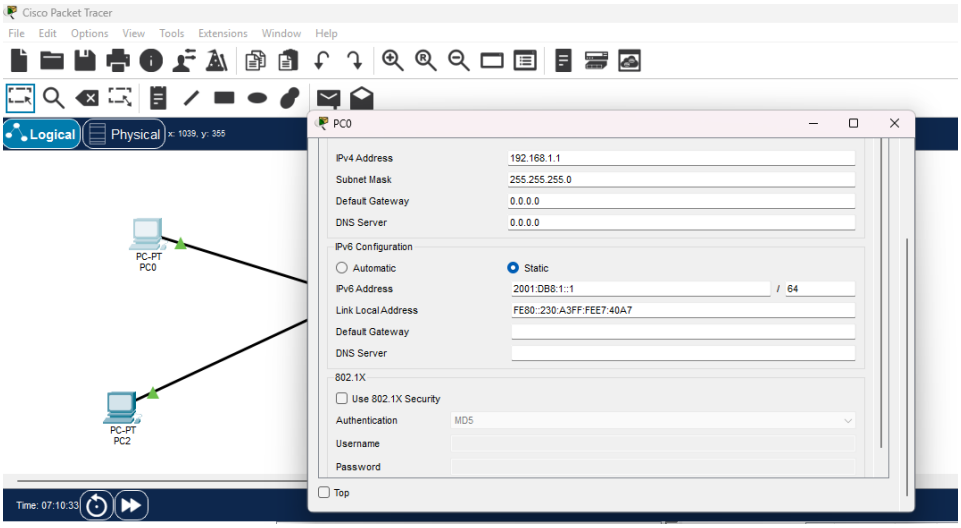

Logical
Physical
x: 323, y: 212
PC2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=16ms TTL=128
Reply from 192.168.1.2: bytes=32 time=14ms TTL=128
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 16ms, Average = 9ms
C:\>
PC-PT
PC0
Hub-PT
Hub0
PC-PT
PC2

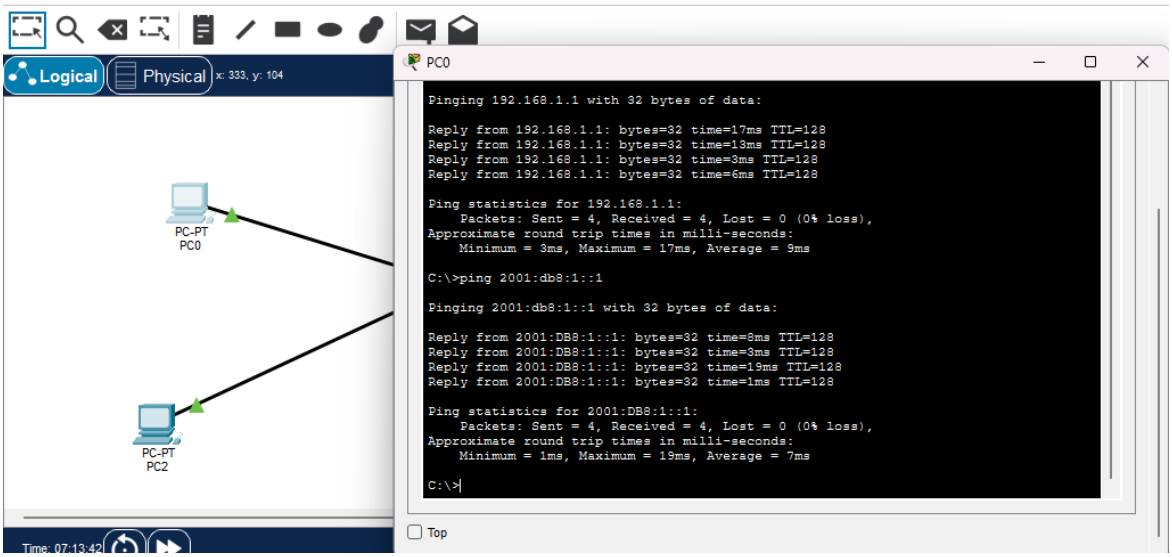
Pc3


PC-PT
PC1
Hub-PT
Hub0
PC-PT
PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static

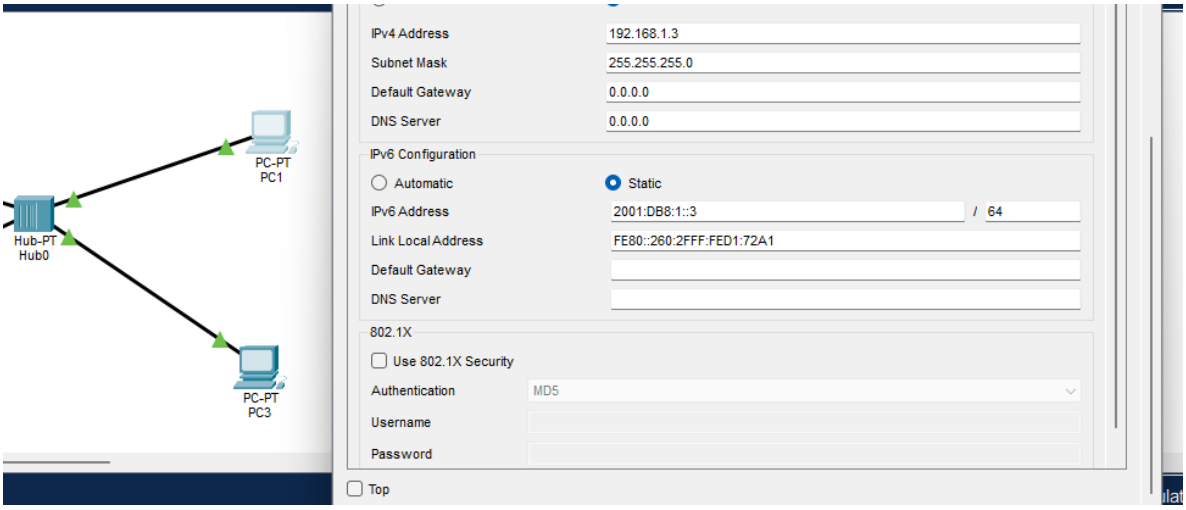


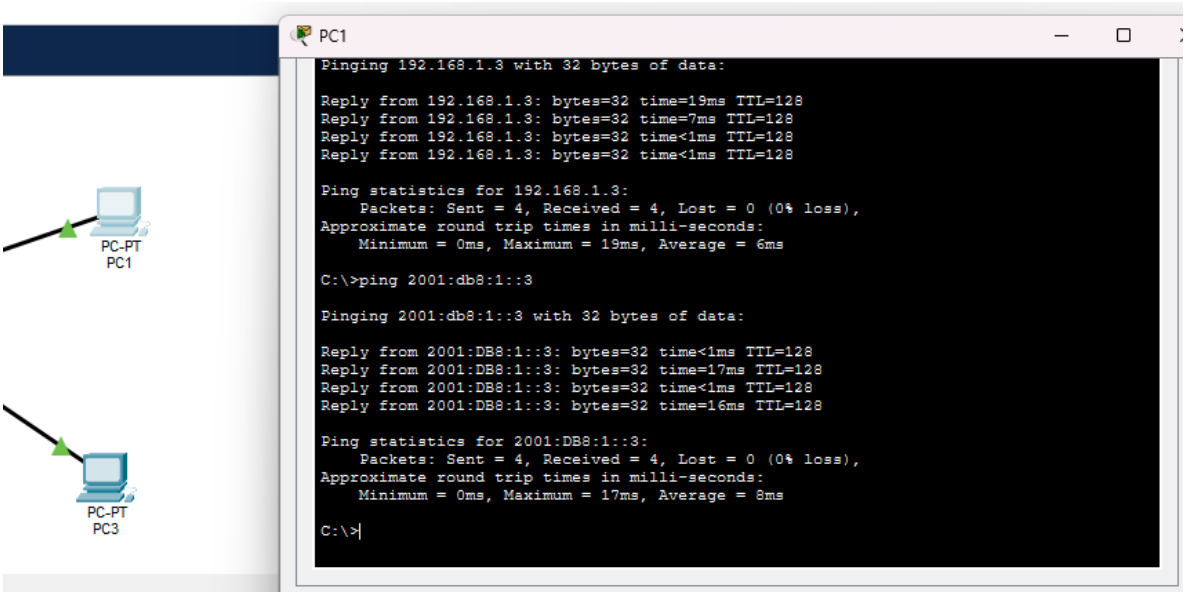
Dans la même topologie, configurez également une adresse IPv6 sur chaque PC :



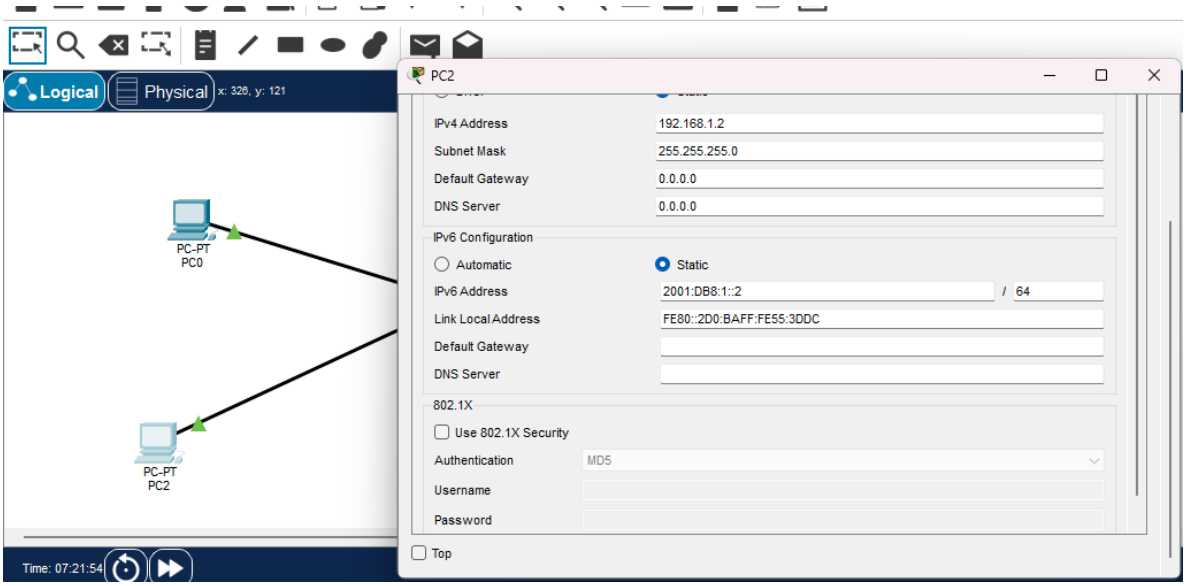


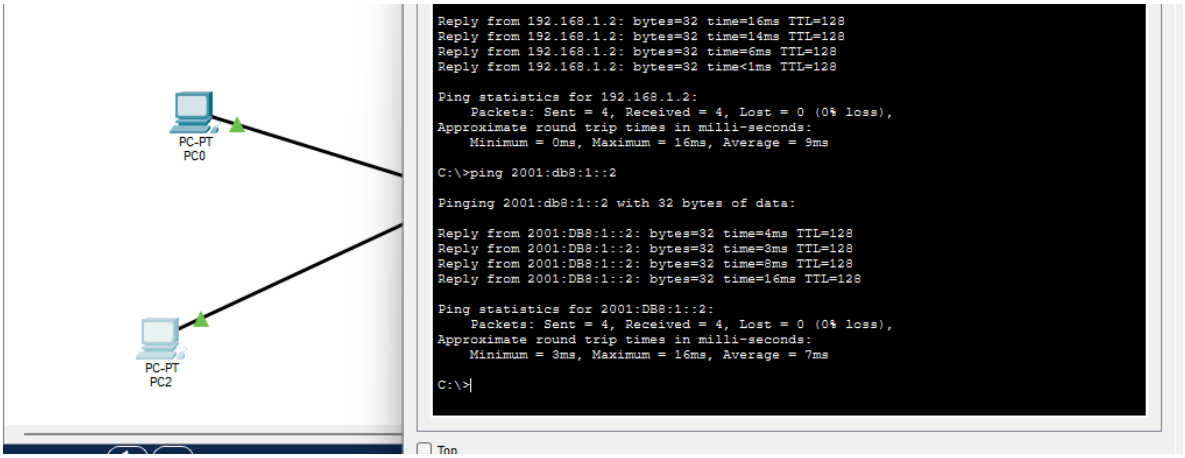
Pc1


PC-PT
PC1
Hub-PT
Hub0
PC-PT
PC3
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::3
/ 64
Link Local Address
FE80::260:2FFF:FED1:72A1
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

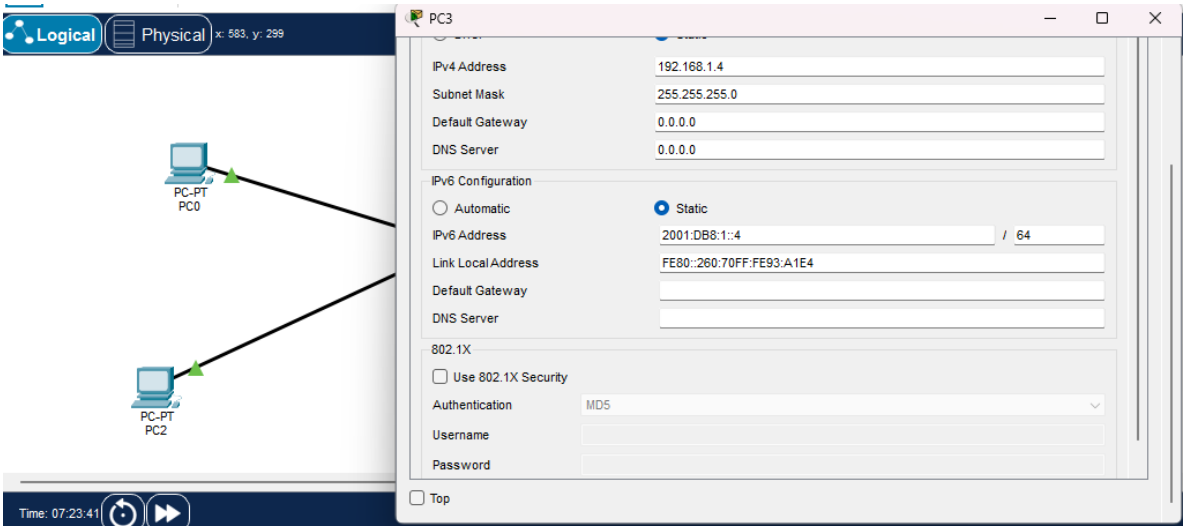

PC-PT
PC1
PC-PT
PC3
PC1
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=19ms TTL=128
Reply from 192.168.1.3: bytes=32 time=7ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 19ms, Average = 6ms
C:\>ping 2001:db8:1::3
Pinging 2001:db8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=17ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=16ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 8ms
C:\>

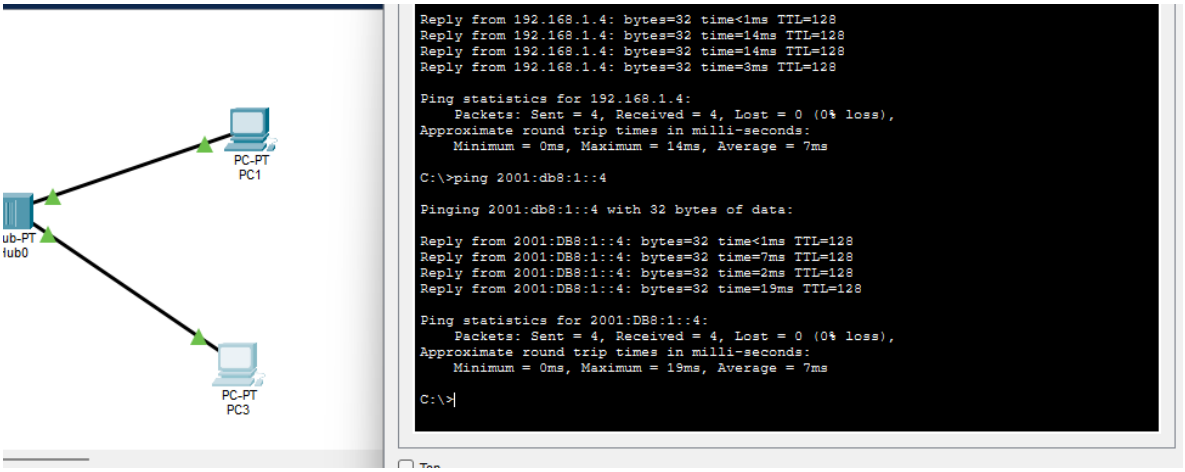
Pc2


Logical
Physical
x: 326, y: 121
PC-PT
PC0
PC-PT
PC2
Time: 07:21:54
PC2
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::2
/ 64
Link Local Address
FE80::2D0:BAFF:FE55:3DDC
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

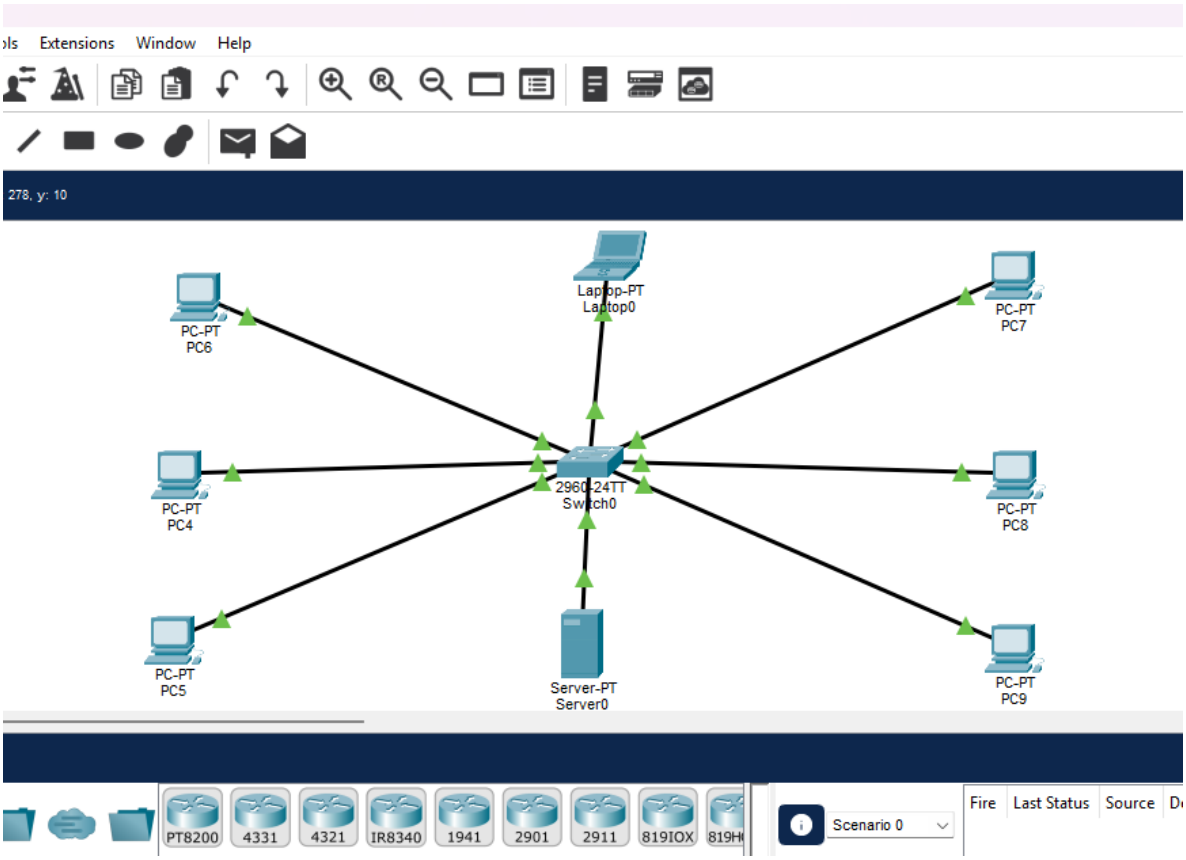

PC-PT
PC0
PC-PT
PC2
Reply from 192.168.1.2: bytes=32 time=16ms TTL=128
Reply from 192.168.1.2: bytes=32 time=14ms TTL=128
Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 16ms, Average = 9ms
C:\>ping 2001:db8:1::2
Pinging 2001:db8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=3ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=8ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=16ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 3ms, Maximum = 16ms, Average = 7ms
C:\>

PC 3

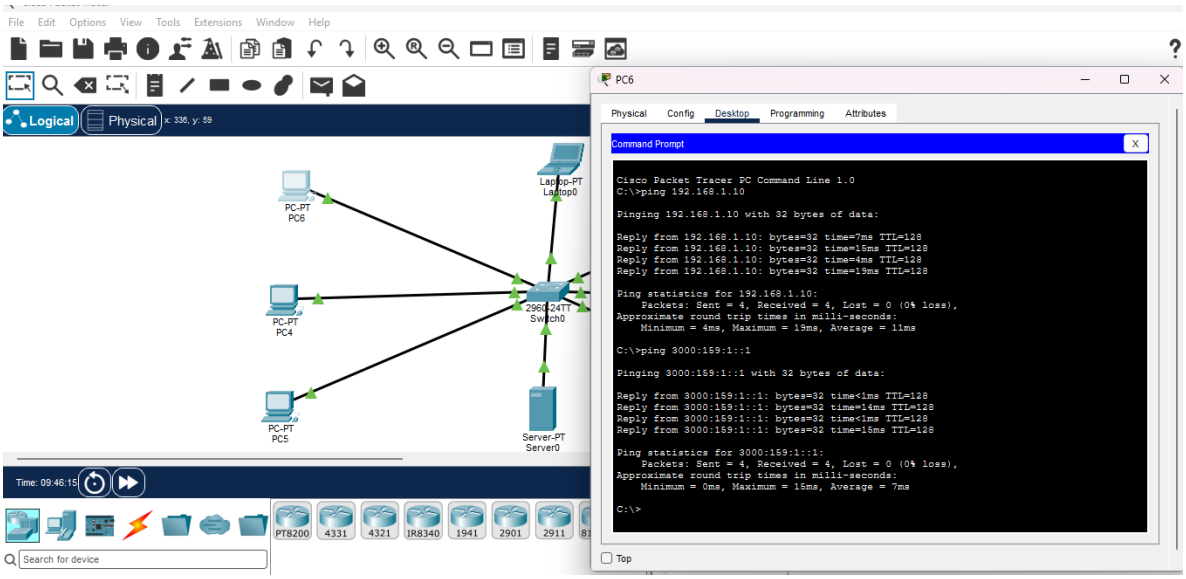

Logical
Physical
x: 583, y: 299
PC-PT
PC0
PC-PT
PC2
Time: 07:23:41
PC3
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::4
/ 64
Link Local Address
FE80::260:70FF:FE93:A1E4
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC-PT
PC1
ub-PT
ub0
PC-PT
PC3
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=14ms TTL=128
Reply from 192.168.1.4: bytes=32 time=14ms TTL=128
Reply from 192.168.1.4: bytes=32 time=3ms TTL=128
Ping statistics for 192.168.1.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 7ms
C:\>ping 2001:db8:1::4
Pinging 2001:db8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=7ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=2ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=19ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 19ms, Average = 7ms
C:\>

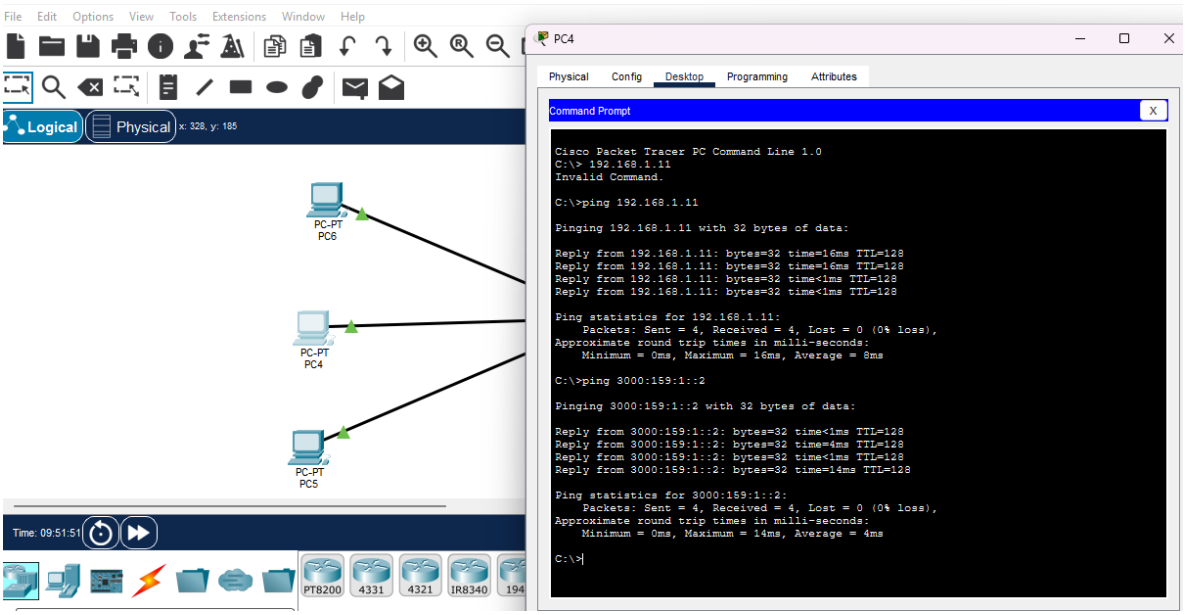
6. Parmi les topologies, choisissez-en une, configurez les adresses ipv4 ivp6



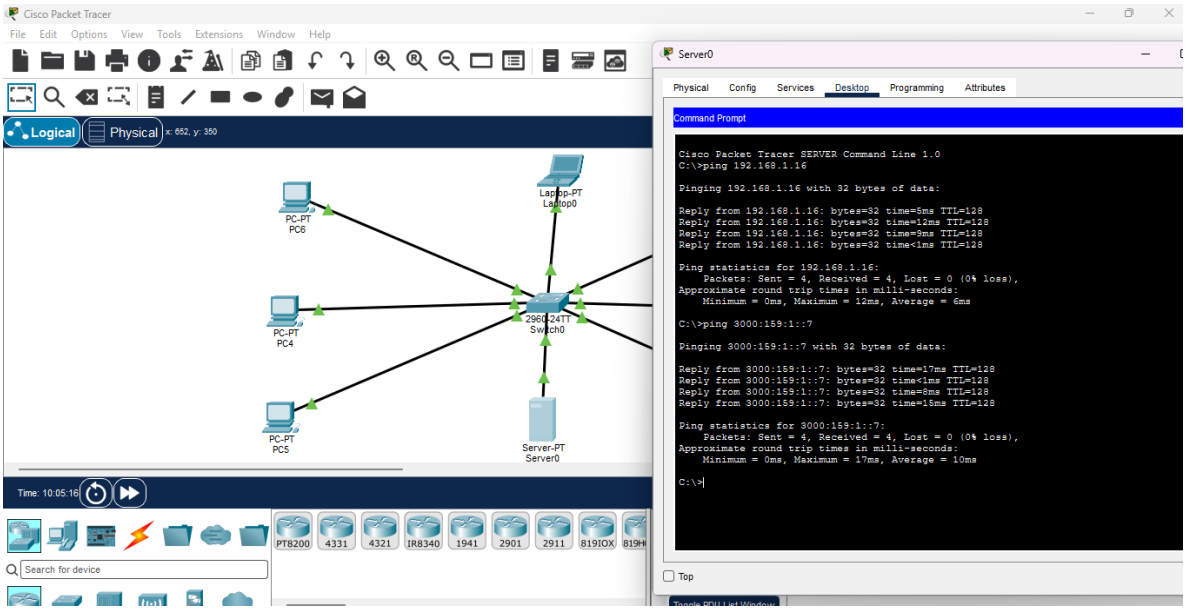
PC6



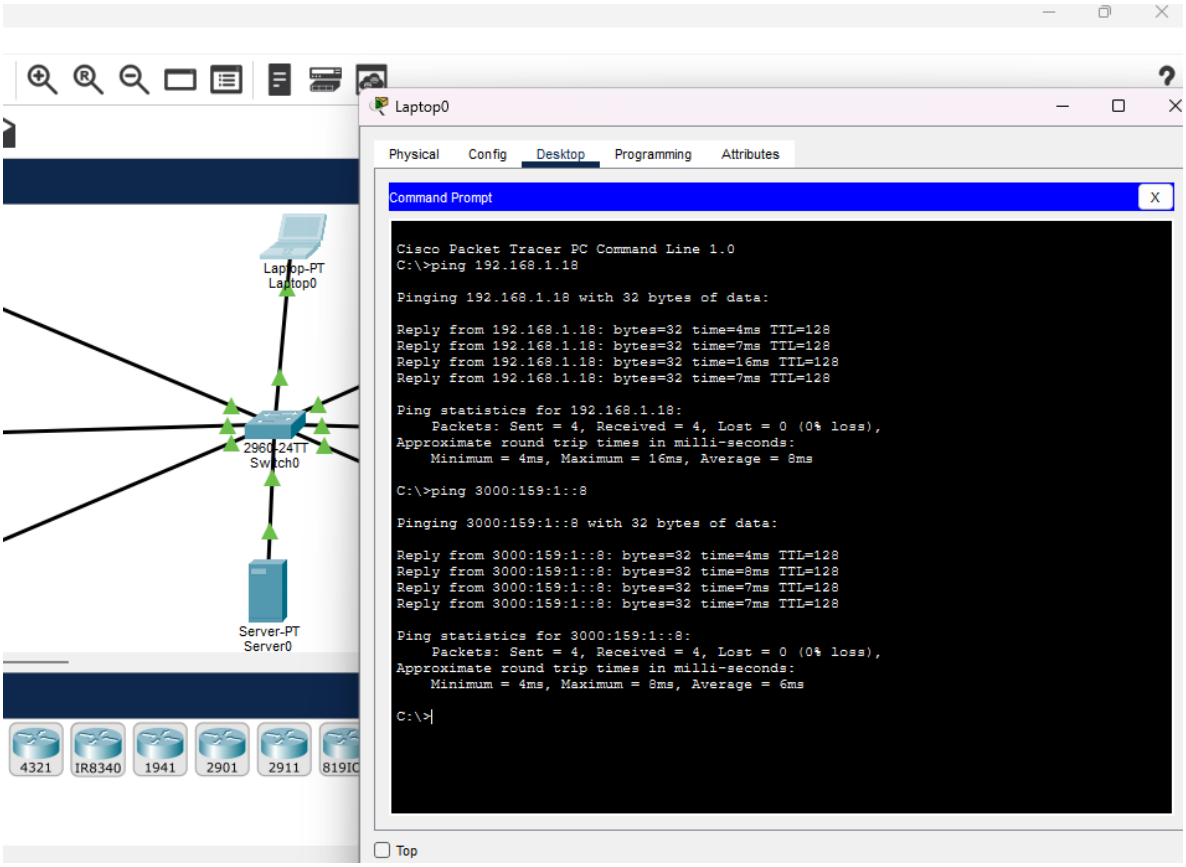
PC 4


File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC6
PC-PT PC4
PC-PT PC5
Time: 09:51:51
PT8200
4331
4321
IR8340
PC4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\> 192.168.1.11
Invalid Command.
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time=16ms TTL=128
Reply from 192.168.1.11: bytes=32 time=16ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 16ms, Average = 8ms
C:\>ping 3000:159:1::2
Pinging 3000:159:1::2 with 32 bytes of data:
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::2: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::2: bytes=32 time=14ms TTL=128
Ping statistics for 3000:159:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 14ms, Average = 4ms
C:\>

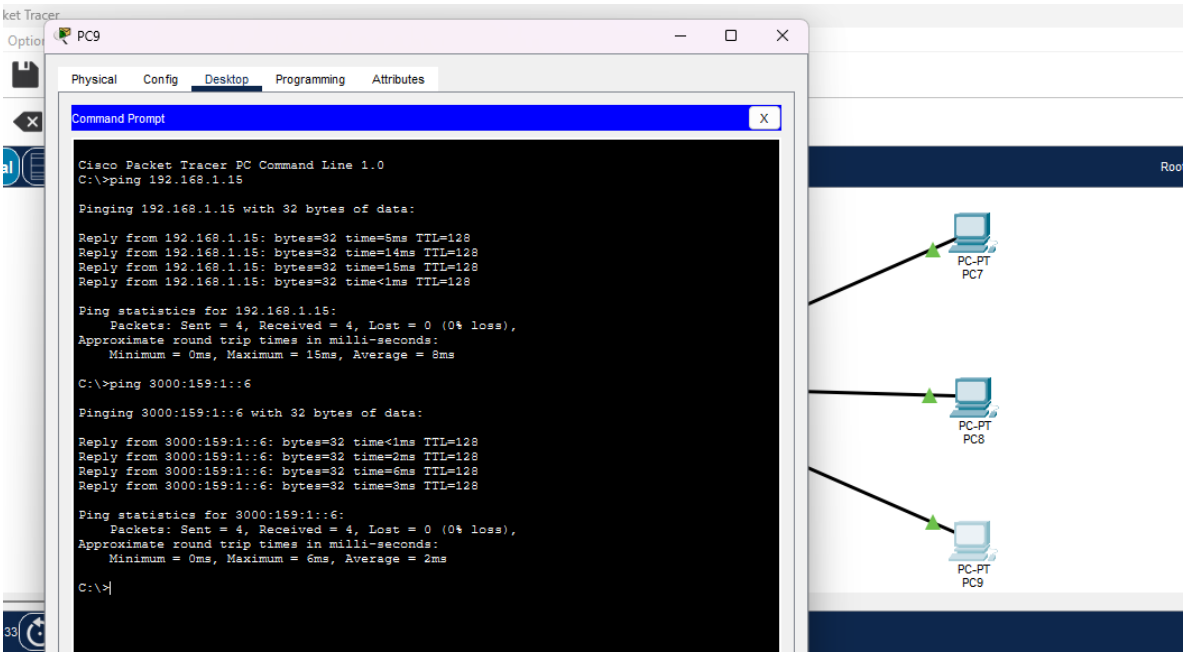
Server0


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC6
PC-PT PC4
PC-PT PC5
Laptop-PT Laptop0
2960-24TT Switch0
Server-PT Server0
Time: 10:05:16
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
Search for device
Server0
Physical Config Services Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time=5ms TTL=128
Reply from 192.168.1.16: bytes=32 time=12ms TTL=128
Reply from 192.168.1.16: bytes=32 time=9ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.16:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 6ms
C:\>ping 3000:159:1::7
Pinging 3000:159:1::7 with 32 bytes of data:
Reply from 3000:159:1::7: bytes=32 time=17ms TTL=128
Reply from 3000:159:1::7: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::7: bytes=32 time=8ms TTL=128
Reply from 3000:159:1::7: bytes=32 time=15ms TTL=128
Ping statistics for 3000:159:1::7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 10ms
C:\>
Top

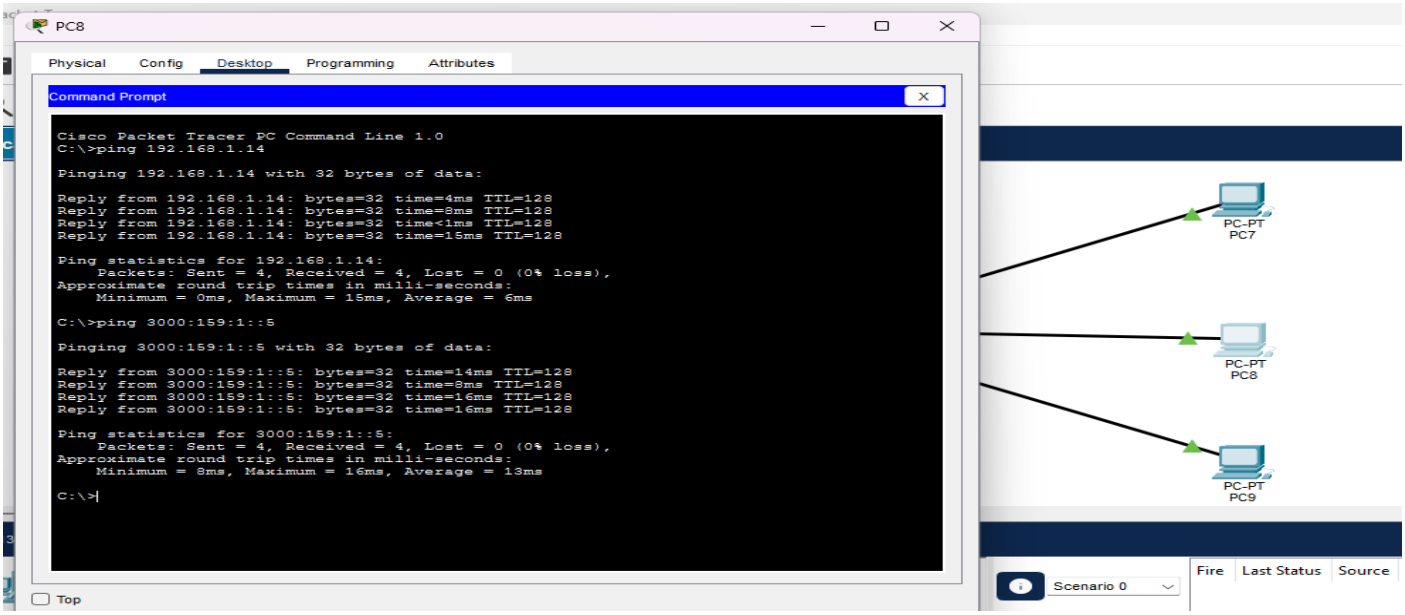
Laptop0


Laptop0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.18
Pinging 192.168.1.18 with 32 bytes of data:
Reply from 192.168.1.18: bytes=32 time=4ms TTL=128
Reply from 192.168.1.18: bytes=32 time=7ms TTL=128
Reply from 192.168.1.18: bytes=32 time=16ms TTL=128
Reply from 192.168.1.18: bytes=32 time=7ms TTL=128
Ping statistics for 192.168.1.18:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 16ms, Average = 8ms
C:\>ping 3000:159:1::8
Pinging 3000:159:1::8 with 32 bytes of data:
Reply from 3000:159:1::8: bytes=32 time=4ms TTL=128
Reply from 3000:159:1::8: bytes=32 time=8ms TTL=128
Reply from 3000:159:1::8: bytes=32 time=7ms TTL=128
Reply from 3000:159:1::8: bytes=32 time=7ms TTL=128
Ping statistics for 3000:159:1::8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 8ms, Average = 6ms
C:\>
Top
Laptop-PT
Laptop0
2960-24TT
Switch0
Server-PT
Server0
4321
IR8340
1941
2901
2911

PC9



PC8



Résumer de ce que j’ai appris.

Aujourd'hui, j'ai réalisé le TP n°3.

Il avait pour premiers objectifs de me faire comprendre ce que sont **IPv4** et **IPv6**, les raisons de leur création ainsi que leurs rôles respectifs.

De plus, ce TP m'a permis d'apprendre à utiliser la commande **Ping**, un outil essentiel pour vérifier si mon ordinateur (PC ou laptop) peut communiquer et se connecter avec d'autres équipements informatiques ou serveurs sur un réseau.

Indiquer si la tâche est réussie, difficultés rencontrées et solutions.

Oui la tâche est réussie pas trop de difficultés pour ce TD